University of Ottawa

Department of Mathematics and Statistics

MAT 1341 : Introduction to Linear Algebra

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Second Midterm Exam – Version A

October 23, 2025

Surname _____ First Name _____

Student # _____ DGD _____

Instructions:

(a) You have 80 minutes to complete this exam.

(b) All work to be considered for grading should be written on your solution sheets. You should clearly indicate where the solution of each question is written.

(c) Write your student number at the top of each page in the space provided.

(d) The last page can be detached and be used as scrap paper. You may also use it as extra space for solutions (in this case, do not detach it).

(e) Cellular phones, unauthorized electronic devices or course notes are not allowed during this exam. Phones and devices must be turned off and put away in your bag. Do not keep them in your possession, such as in your pockets. If caught with such a device or document, the following may occur: you will be asked to leave immediately the exam and academic fraud allegations will be filed which may result in you obtaining a 0 (zero) for the exam.

(f) You are strongly recommended to write in **pen**, not pencil.

By signing below, you acknowledge that you are complying with statement (h) above:

Signature: _____

Please do not write in the table below.

Question	1	2	3	4	5	Total
Maximum	2	5	4	4	3	18
Grade						

1. **[2 points]** For each statement below, write '**T**' if the statement is true, and write '**F**' if the statement is false. You will receive .5 points for each correct answer. There are no negative points.

____ If A is a 6×7 matrix such that $\text{rank}(A) = 5$, then the columns of A are linearly dependent vectors.

____ If $H = \text{Span}\{\mathbf{a}, \mathbf{b}, \mathbf{c}\}$ for 3 nonzero vectors $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^5$, then $\dim(H) = 3$.

____ If S is a linearly dependent subset of $\mathbb{R}^5$, then every non-empty subset of S must also be linearly dependent.

____ The solutions of the homogeneous linear system $A\mathbf{x} = \mathbf{0}$ where

$$A = \begin{bmatrix} 3 & -4 & 2 & 1 & 7 \\ 0 & 0 & 1 & 1 & 1 \\ -4 & 1 & 0 & -5 & 3 \end{bmatrix}$$

form a subspace of $\mathbb{R}^3$.

Solution: T, F, F, F

2. Let H be the subspace of $\mathbb{R}^4$ defined by

$$H = \left\{ \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \;\middle|\; x + z = y + 2w \right\}.$$

You do not need to verify that H is a subspace.

(a) **[3 points]** Find a basis for H. You should justify that the list of vectors that you find form a basis. **Solution:** We have $x = y - z + 2w$. Making a substitution, we have

$$\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} y - z + 2w \\ y \\ z \\ w \end{bmatrix} = y \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + z \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + w \begin{bmatrix} 2 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Since y, z, w are free variables (and only x is basic), the vectors

$$\begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

form a spanning set of H. But they are also linearly independent. There at at least 3 ways to see this:

(I) Because the matrix formed by the vectors is

$$\begin{bmatrix} 1 & -1 & 2 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

and the reduced echelon form of this matrix is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

thus every column is a pivot column, this the vectors are linearly independent.

(II) H is the null space of the matrix $\begin{bmatrix} 1 & -1 & 1 & -2 \end{bmatrix}$. The above procedure is the method of finding a basis for the null space.

(III) If we have

$$y \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + z \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + w \begin{bmatrix} 2 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

for some scalars x, y, z, then we have

$$\begin{bmatrix} y - z + 2w \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

and by equating the entries we obtain $x = y = z = 0$. Thus, the vectors are linearly independent.

(b) **[2 points]** Set $\mathbf{w} = \begin{bmatrix} 7 \\ -11 \\ 12 \\ 3 \end{bmatrix}$ and let

$$K = \{\mathbf{v} \in H \ : \ \mathbf{v} \bullet \mathbf{w} = 0\}.$$

Here $\mathbf{v} \bullet \mathbf{w}$ means the dot product of $\mathbf{v}$ and $\mathbf{w}$. Is K a subspace of H? You should justify your answer.

Solution: Yes, we have $\mathbf{0} \bullet \mathbf{w} = 0$, hence $\mathbf{0} \in K$. Also

$$\mathbf{v}_1, \mathbf{v}_2 \in K \Rightarrow \mathbf{v}_1 \bullet \mathbf{w} = 0 \text{ and } \mathbf{v}_2 \bullet w = 0.$$

Thus

$$(\mathbf{v}_1 + \mathbf{v}_2) \bullet \mathbf{w} = \mathbf{v}_1 \bullet \mathbf{w} + \mathbf{v}_2 \bullet \mathbf{w} = 0 + 0 = 0.$$

This proves $\mathbf{v}_1 + \mathbf{v}_2 \in K$.

Similarly, for $\mathbf{v} \in K$ we have

$$\mathbf{v} \in K \Rightarrow \mathbf{v} \bullet \mathbf{w} = 0 \Rightarrow (c\mathbf{v}) \bullet \mathbf{w} = c(\mathbf{v} \bullet \mathbf{w}) = c(0) = 0.$$

This proves that $c\mathbf{v} \in K$ as well.

3. Let $A = \begin{bmatrix} 2 & -1 & 3 & 4 & 5 \\ 0 & 0 & -4 & 1 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} & 3 \end{bmatrix}$.

(a) [**2 points**] Find a basis for the vector space Col(A). **Solution:** The matrix is in echelon form and we need to take the pivot columns. Thus, the basis of Col(A) is

$$\left\{ \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ -4 \\ 0 \end{bmatrix}, \begin{bmatrix} 4 \\ 1 \\ -\frac{1}{2} \end{bmatrix} \right\}$$

(b) [**1 point**] Is the vector $\begin{bmatrix} \frac{1}{2} \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ in Nul(A)? You should justify your answer.

Solution: By direct computaion we have

$$\begin{bmatrix} 2 & -1 & 3 & 4 & 5 \\ 0 & 0 & -4 & 1 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} & 3 \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

hence this vector is in Nul(A).

(c) [**1 point**] What is the dimension of Nul(A)? You should justify your answer.

Solution: The dimension of Nul(A) is the number of non-pivot columns of A, i.e., 5-3=2.

4. [**4 points**] Determine if $x-x^3$ is in Span $\left\{x-x^2, 1-2x^3, \frac{1}{2}-x^2\right\}$. You should justify your answer.

Solution: Suppose that

$$x-x^3 = x_1(x-x^2)+x_2(1-2x^3)+x_3(\frac{1}{2}-x^2) = (x_2+\frac{1}{2}x_3)+x_1x+(-x_1-x_3)x^2+(-2x_2)x^3.$$

Equating coefficients of powers of x on both sides, we obtain the linear system

$$\begin{cases} x_2 + \frac{1}{2}x_3 & = 0 \\ x_1 & = 1 \\ -x_1 - x_3 & = 0 \\ -2x_2 & = -1. \end{cases}$$

This linear system can be solved by the row reduction method, but there is a shortcut: from the above equations it follows that

$$x_1 = 1,\ x_2 = \frac{1}{2},\ x_3 = -x_1 = -1.$$

These values are consistent with the first equation: $x_2 + \frac{1}{2}x_3 = 0$. Thus the linear system has a solution, i.e., $x - x^3$ is in Span $\left\{x - x^2, 1 - 2x^3, \frac{1}{2} - x^2\right\}$. Indeed we have

$$x - x^3 = (x - x^2) + \frac{1}{2}(1 - 2x^3) + (-1)(\frac{1}{2} - x^2).$$

5. **[3 points]** Determine all values of the paremeter t for which the vectors

$$\begin{bmatrix}1\\0\\1\end{bmatrix}, \begin{bmatrix}-1\\3\\2\end{bmatrix}, \begin{bmatrix}2\\1\\t+2\end{bmatrix}$$

are linearly independent. You should justify your answer.

Solution: We form a matrix by the vectors, and reduce it to an echelon form:

$$\begin{bmatrix}1 & -1 & 2\\0 & 3 & 1\\1 & 2 & t+2\end{bmatrix} \xrightarrow{R_3\to R_3-R_1} \begin{bmatrix}1 & -1 & 2\\0 & 3 & 1\\0 & 3 & t\end{bmatrix} \xrightarrow{R_3\to R_3-R_2} \begin{bmatrix}1 & -1 & 2\\0 & 3 & 1\\0 & 0 & t-1\end{bmatrix}$$

This matrix is in echelon form. If $t = 1$ then the last column will not be a pivot column. But if $t \neq 1$ then every column is a pivot column. Thus, the vectors are linearly independent for all values $t \neq 1$.